

APPLIED ANATOMY OF UL-III- ARM & PROXIMAL FORARM

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subjects will be dealt

□ Humeral fractures

Surgical Neck Fracture

Midshaft Humeral Fracture

Medial Epicondylar Fracture

pediatric supracondylar fracture

□ Anatomical radiology of elbow region

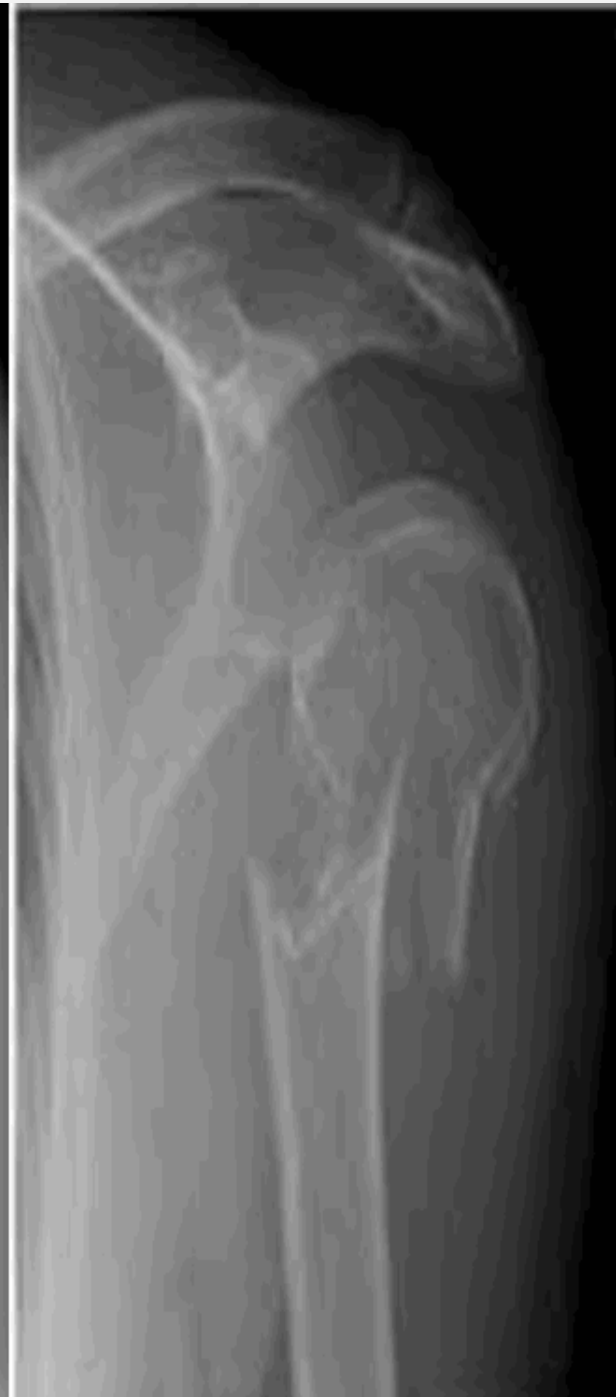
Bony Landmarks

Secondary Centers Of Ossifications

Radiological Lines

Relations of nerves & arteries in the elbow region

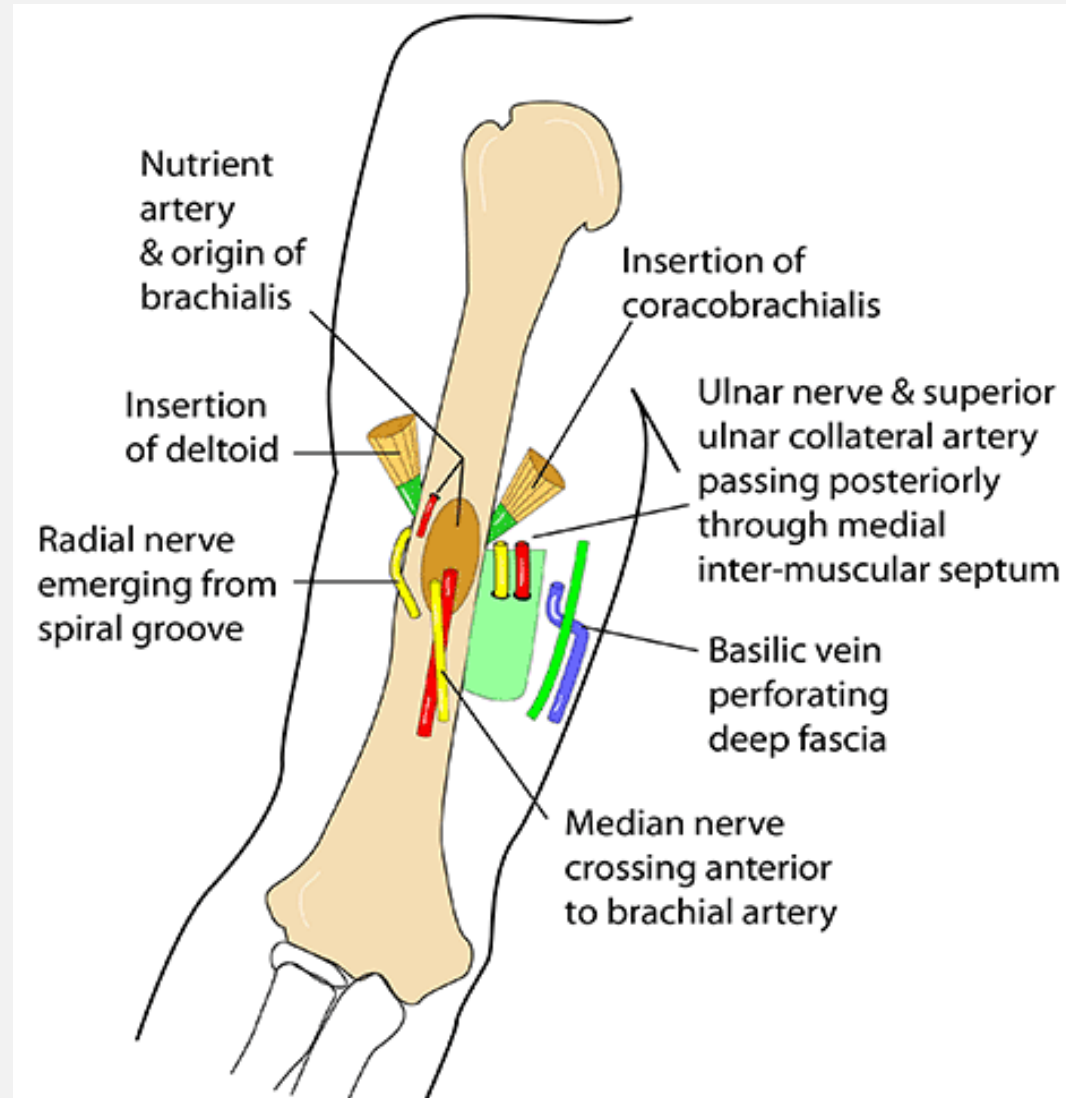
□ Radial Head Subluxation (Nursemaid's Elbow)





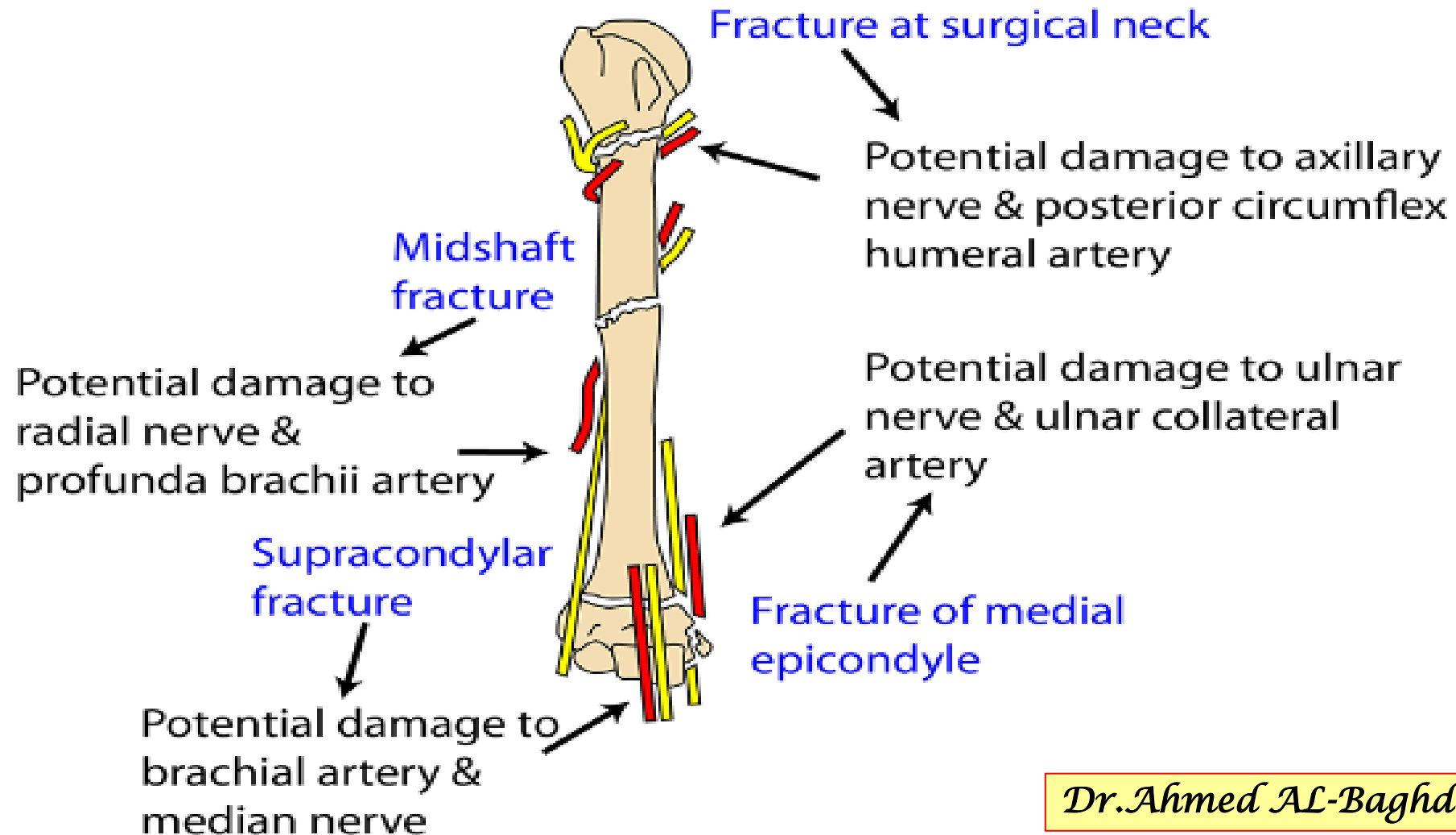
□ what are the anatomical events that occur at or near mid-shaft of humerus?

- Nutrient artery entering humerus
- Origin of brachialis
- Insertion of deltoid
- Insertion of coracobrachialis
- Radial nerve emerging from spiral groove
- Basilic vein perforating deep fascia
- Median nerve crossing anterior to brachial artery
- Ulnar nerve & superior ulnar collateral artery passing posteriorly through medial inter-muscular septum



□ What are the structures in direct contact with humerus and liable for injury?

ARTERIES AND NERVES LIABLE TO DAMAGE AT 4 SITES OF FRACTURE OF THE HUMERUS



Surgical Neck Fracture

Surgical Neck Fracture of Humerus

Case scenario: A 70-year-old woman presents after a **fall onto her outstretched hand** with pain and inability to lift her arm. Exam reveals ecchymosis over the proximal arm, crepitus on motion, and **intact radial pulse**

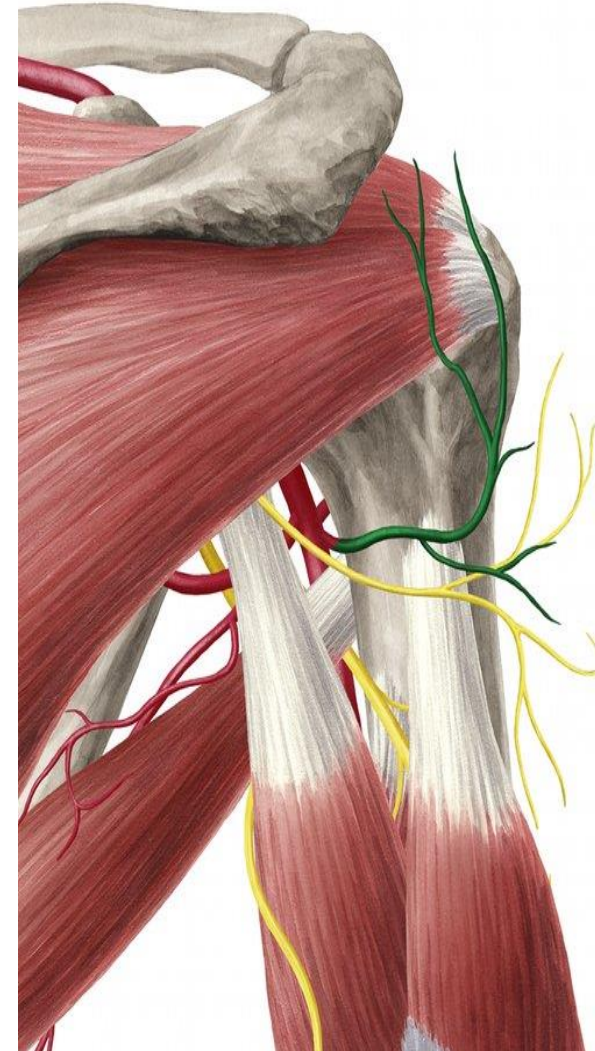


Surgical Neck Fracture of Humerus

Key Features:

- **High-risk population:** Elderly (osteoporosis), seizures, electrocution.
- **Critical structures at risk:**
 - Axillary nerve (runs 5 cm distal to acromion).
 - Posterior circumflex humeral artery (adjacent to nerve).
- **Deformity:** Proximal fragment abducted/externally rotated (pectoralis major pull)

Why "surgical": High risk for neurovascular injury (vs. "anatomical" neck).



Surgical Neck Fracture of Humerus

Q. Which clinical sign is most likely to be found in a patient with a surgical neck fracture of the humerus complicated by nerve injury?

- a) Wrist drop
- b) Loss of shoulder abduction beyond 15°
- c) Loss of elbow flexion
- d) Claw hand deformity
- e) Loss of thumb opposition

Answer:  b) Loss of shoulder abduction beyond 15°

Explanation:

- Surgical neck fractures put the **axillary nerve** at risk, as it winds around the neck with the posterior circumflex humeral artery.
- The axillary nerve supplies the **deltoid and teres minor**.
 - → Loss of **shoulder abduction (15°–110°)** due to deltoid paralysis.
 - → Loss of shoulder contour (deltoid wasting).
 - → Sensory loss over the “regimental badge” area (upper lateral arm).

Surgical Neck Fracture of Humerus

Q. Which movement is MOST affected in axillary nerve injury?

- a) Elbow flexion
- b) Shoulder abduction beyond 15°
- c) Wrist extension

Answer: ☒ b

Explanation: Deltoid (axillary nerve) is primary abductor from 15° – 110° . Supraspinatus initiates first 15°

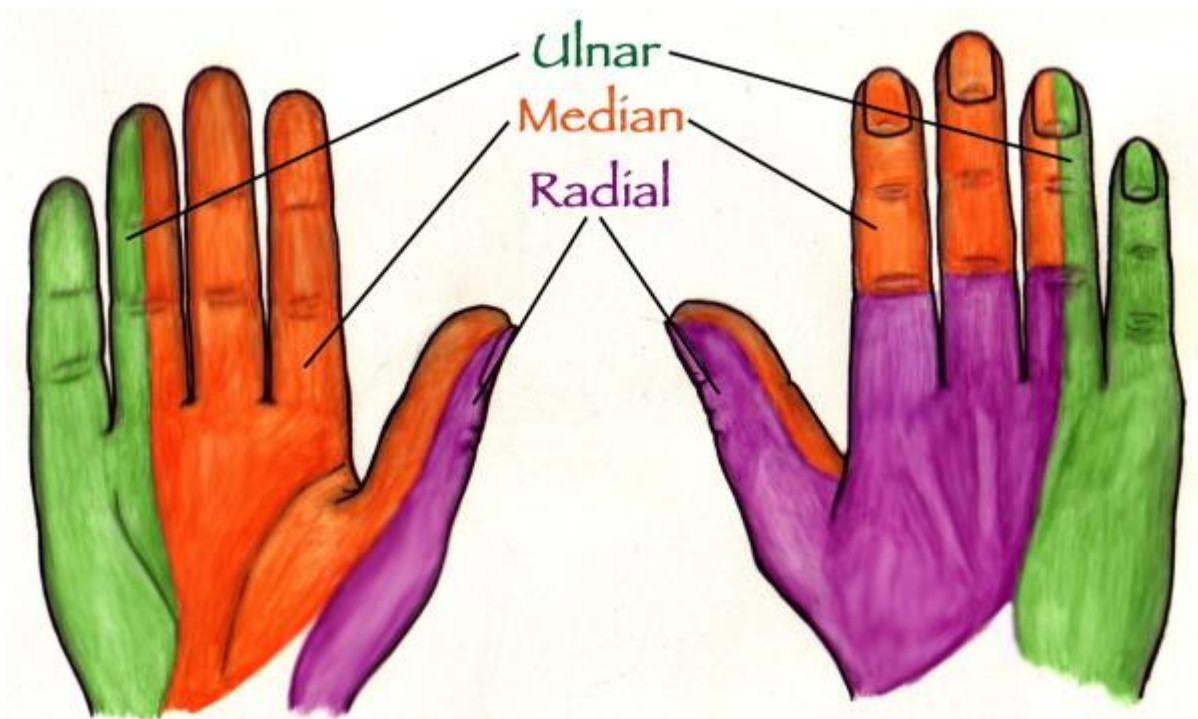
Q: displacement

Q: risk of injury

Midshaft Humeral Fracture

Midshaft Humeral Fracture

Case scenario: 35-year-old construction worker falls from scaffolding, landing on his right arm. He presents with **visible deformity at mid-arm**, **weak wrist extension**, and **numbness over the dorsum of the hand**.



Midshaft Fracture of Humerus

Key Features:

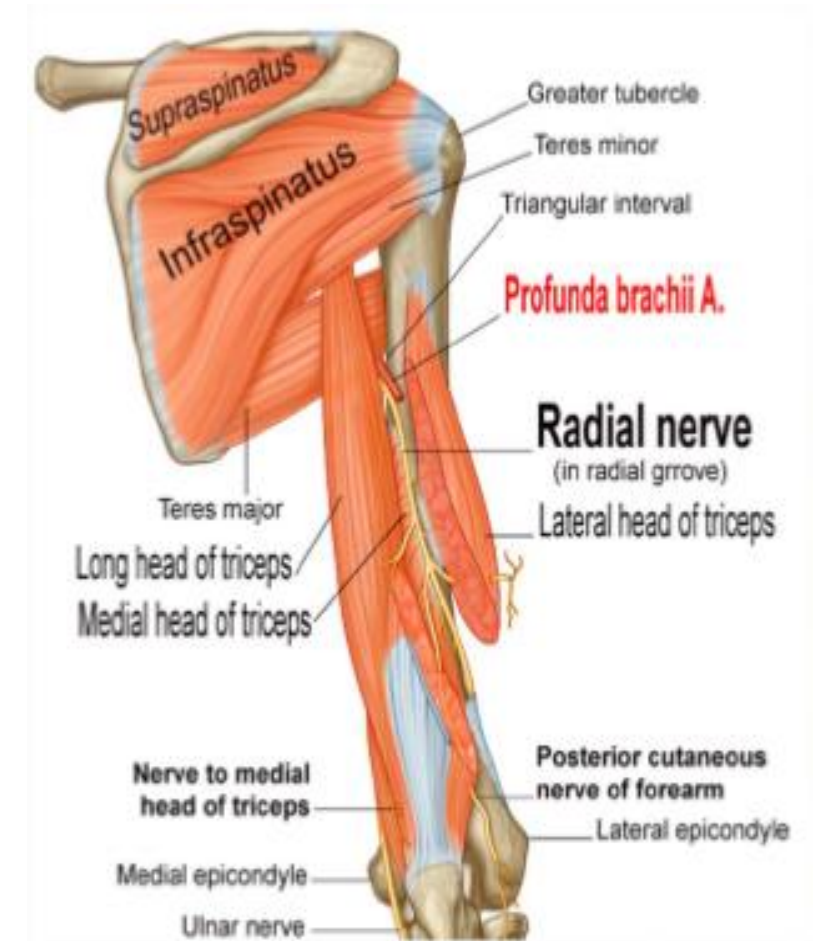
- **"Hanging arm" deformity**: Dependent distal fragment due to gravity.
- **Radial nerve injury**: Wraps around spiral groove (10-14 cm distal to acromion)

Anatomical features

1. Radial Nerve Course:

- **Proximal to fracture**: Innervates triceps (often spared in midshaft fractures).
- **At spiral groove**: Vulnerable to laceration/stretch (test: wrist/finger extension).
- **Distal to fracture**: Emerges lateral to brachioradialis (sensory: dorsal hand). ??

2. Muscle Forces & Displacement (depend on site of fractures) ??

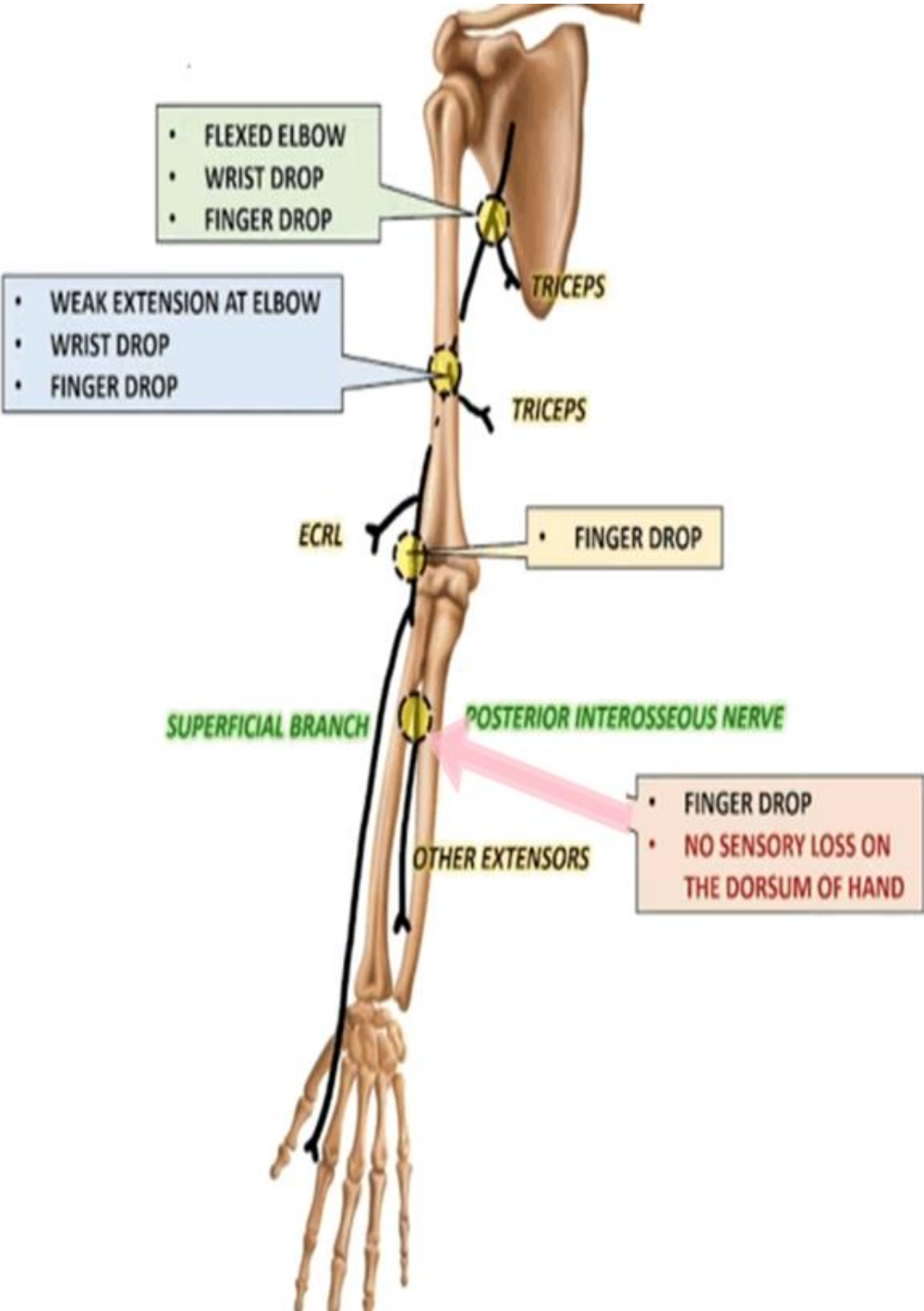
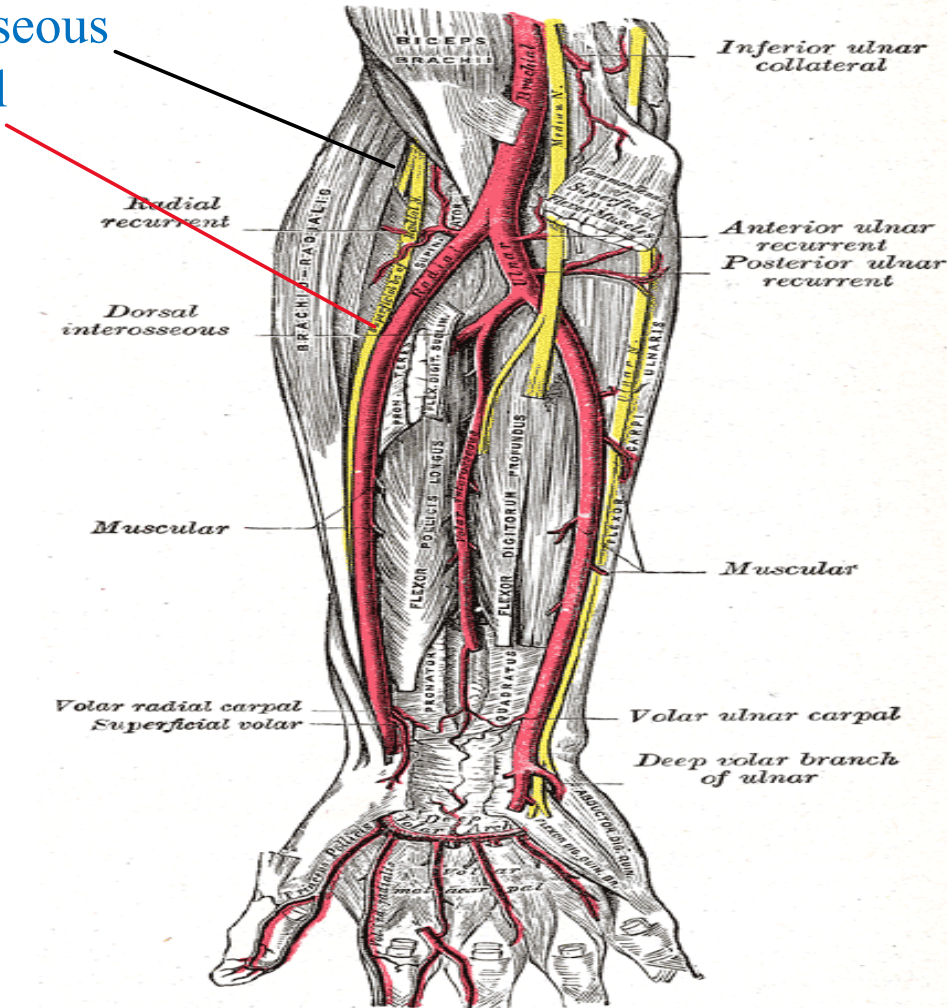


 in fractures of humeral shaft, the fractured segments displaced depends on sit of fracture. Explain how is that?

Site of fracture	Proximal segment	Distal segment
Above bicipital groove	Abducted by muscles attached to greater tuberosity(mainly supraspinatus)	Adducted by muscles attached to bicipital groove
Above the level of deltoid tuberosity	Adducted by muscles attached to bicipital groove	Abducted & pulled up by deltoid muscle
Below the level of deltoid tuberosity	Abducted by deltoid	Adducted & pulled up by biceps & triceps

Diagram shows the common sites of radial nerve injuries & their features

Posterior interosseous
Superficial radial



Midshaft Fracture

Q. Which clinical finding is MOST characteristic of radial nerve injury at the spiral groove?

- a) Wrist drop
- b) Claw hand
- c) Thenar wasting
- d) Ape hand

Answer:  a

Explanation: Wrist drop occurs due to paralysis of wrist and finger extensors supplied by radial nerve.

Midshaft Fracture

Q. Which artery accompanies the radial nerve in the spiral groove of humerus?

- a) Brachial artery
- b) Profunda brachii artery
- c) Radial recurrent artery
- d) Superior ulnar collateral

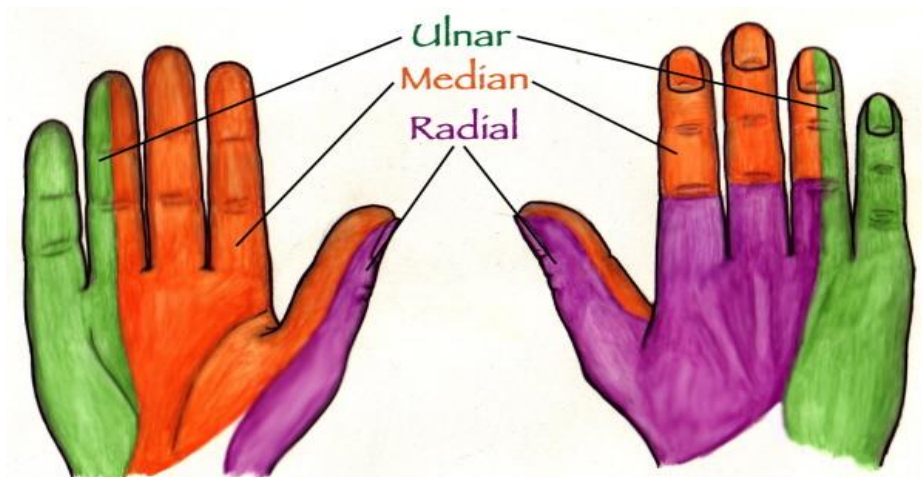
Answer:  b

Explanation: Profunda brachii artery runs with radial nerve in the spiral groove → both can be injured in midshaft fracture.

Medial Epicondylar Fracture of Humerus

Medial Epicondylar Fracture

Case scenario: A 12-year-old boy fell on his **outstretched hand** while playing football. He presents with pain, swelling, and **tenderness over the medial** side of the right elbow. On examination, there is weakness of finger **abduction/adduction and tingling over the ring and little fingers**. Radiograph shows a displaced medial epicondylar fracture of the humerus.



Widening of the Medial Epicondyle Physis

Contralateral Comparison View Without Widening



Medial Epicondylar Fracture

Epidemiology: Common in children and adolescents; accounts for ~10–20% of pediatric elbow fractures.

Mechanism of Injury:

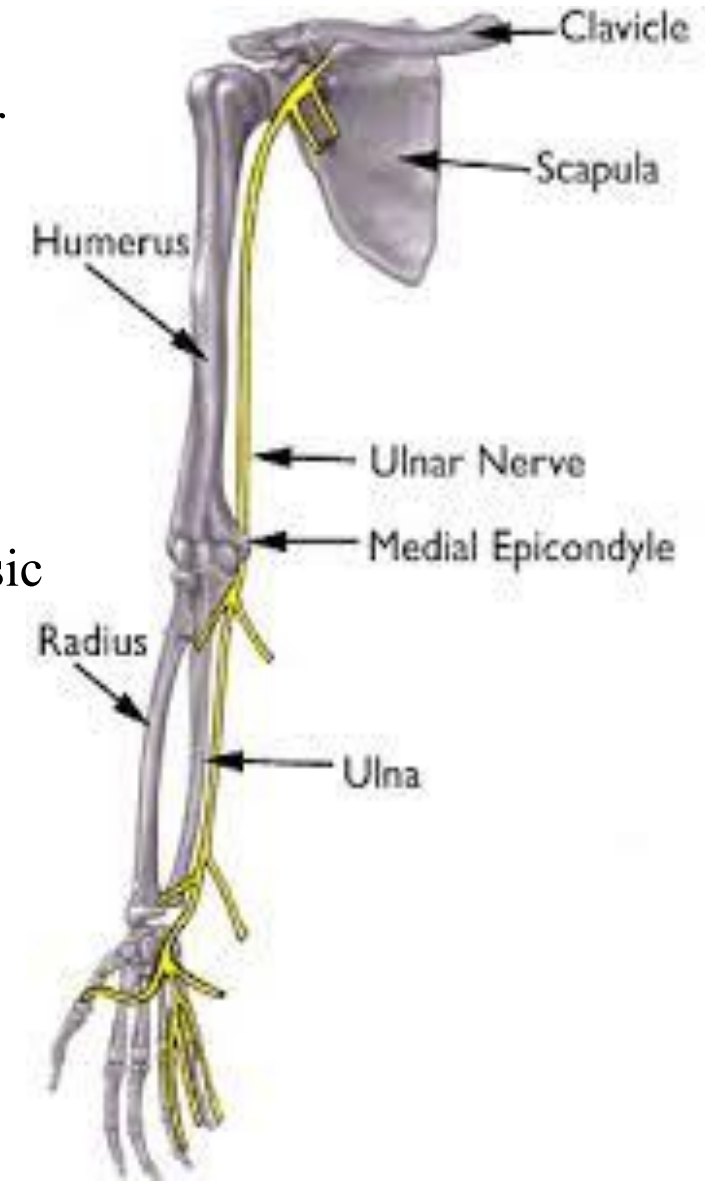
- Fall on an outstretched hand (avulsion of medial epicondyle by flexor-pronator muscles).
- Direct blow to medial elbow.
- Elbow dislocation associated in up to 50%.

Clinical Features:

- Medial elbow pain and swelling
- Tenderness over medial epicondyle.
- Ulnar nerve symptoms: paresthesia in 4th & 5th fingers, clawing, weak intrinsic hand muscles.

Complications:

- Ulnar nerve palsy.
- Elbow instability (due to medial collateral ligament attachment).
- Nonunion of fragment.



Medial Epicondylar Fracture

Anatomical Focus

- Medial epicondyle of humerus: attachment for flexor-pronator group (flexor carpi ulnaris, pronator teres, flexor digitorum superficialis, etc.) and ulnar collateral ligament.
- Ulnar nerve (C8–T1): passes posterior to medial epicondyle in the cubital tunnel. Easily stretched/entrapped in fracture.
- Clinical deficit in ulnar nerve injury:
 - a) **Motor**: intrinsic hand muscle weakness → partial claw hand (ring & little fingers), loss of finger abduction/adduction, weak grip.
 - b) **Sensory**: medial 1½ fingers and hypothenar eminence.

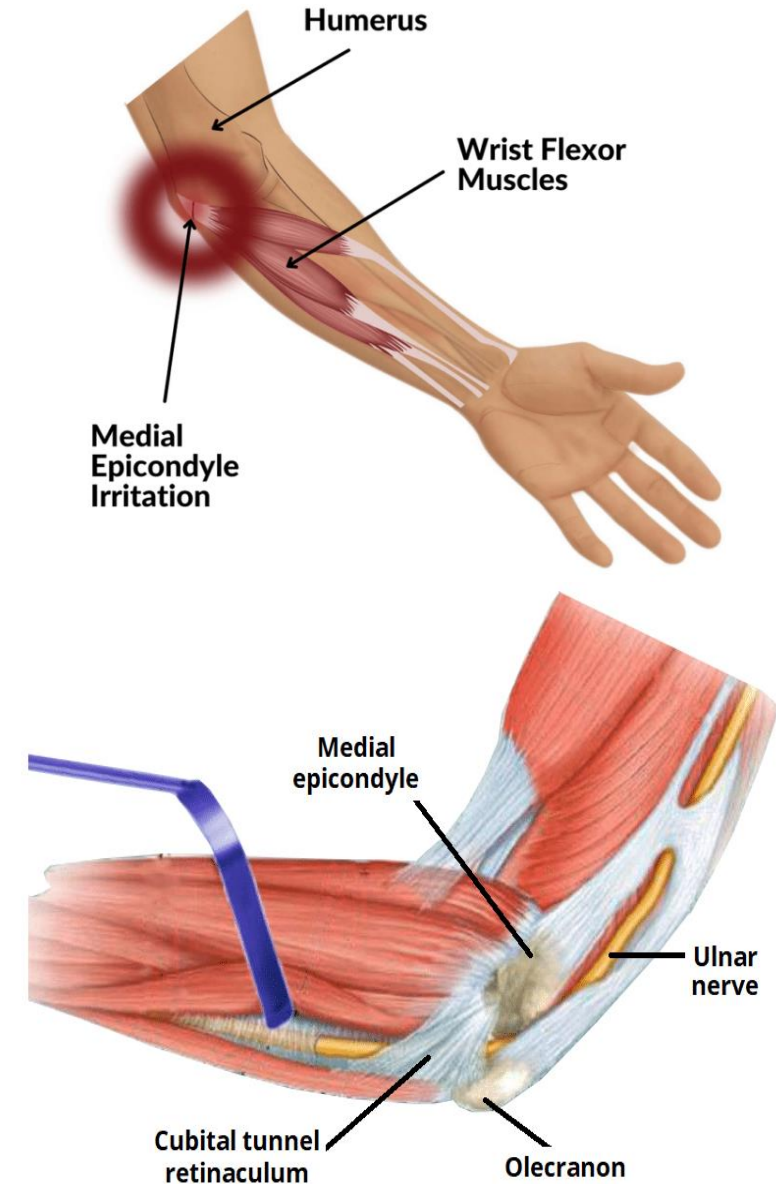
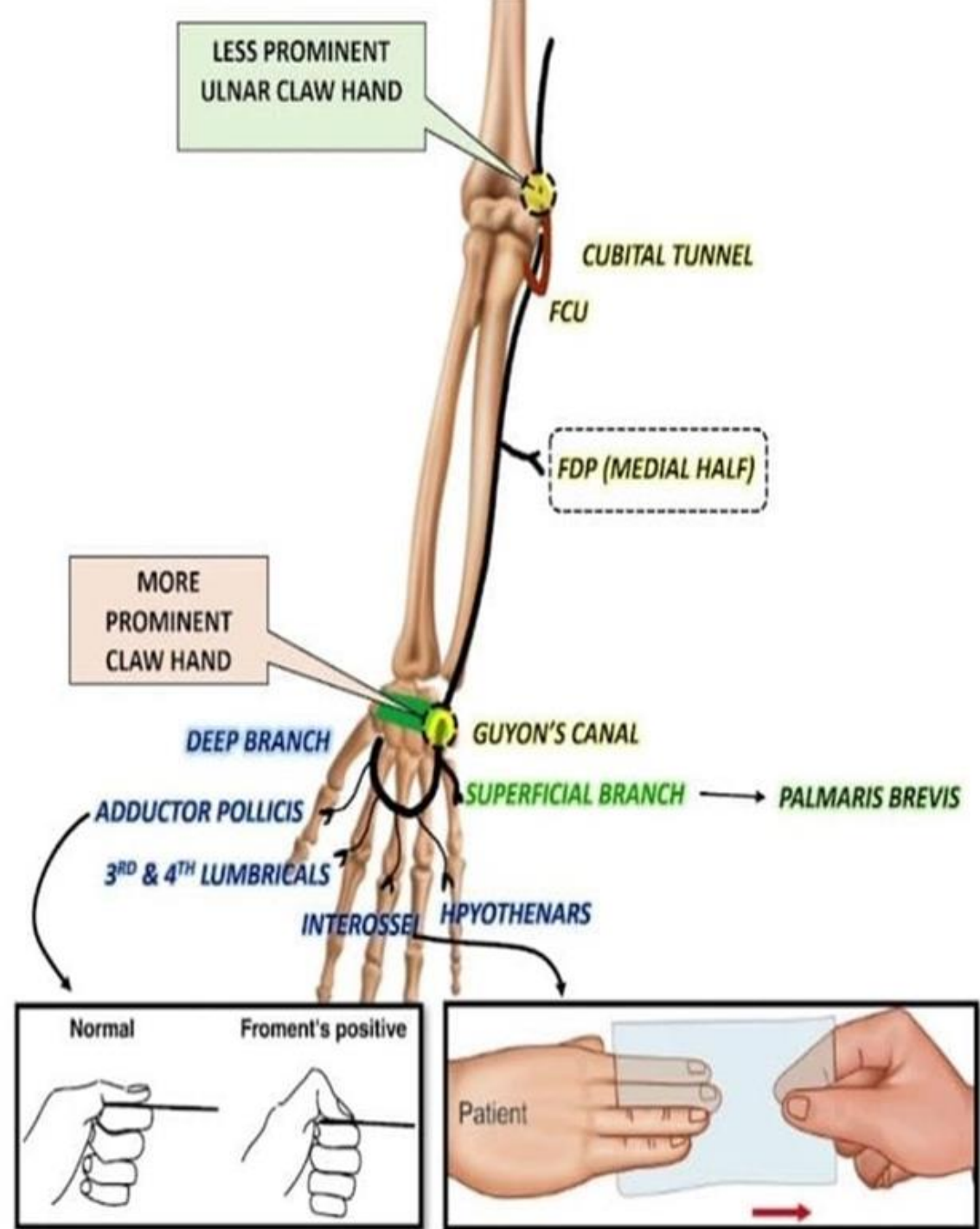
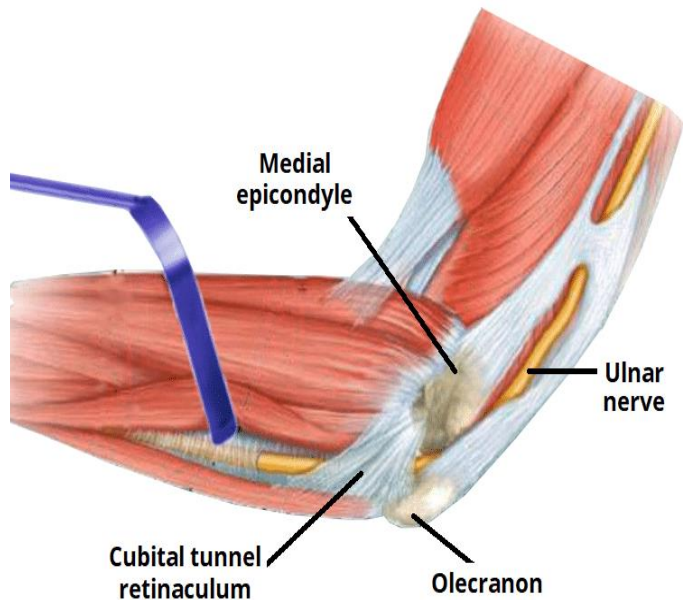


Diagram shows the common sites of Ulnar nerve injuries & their features



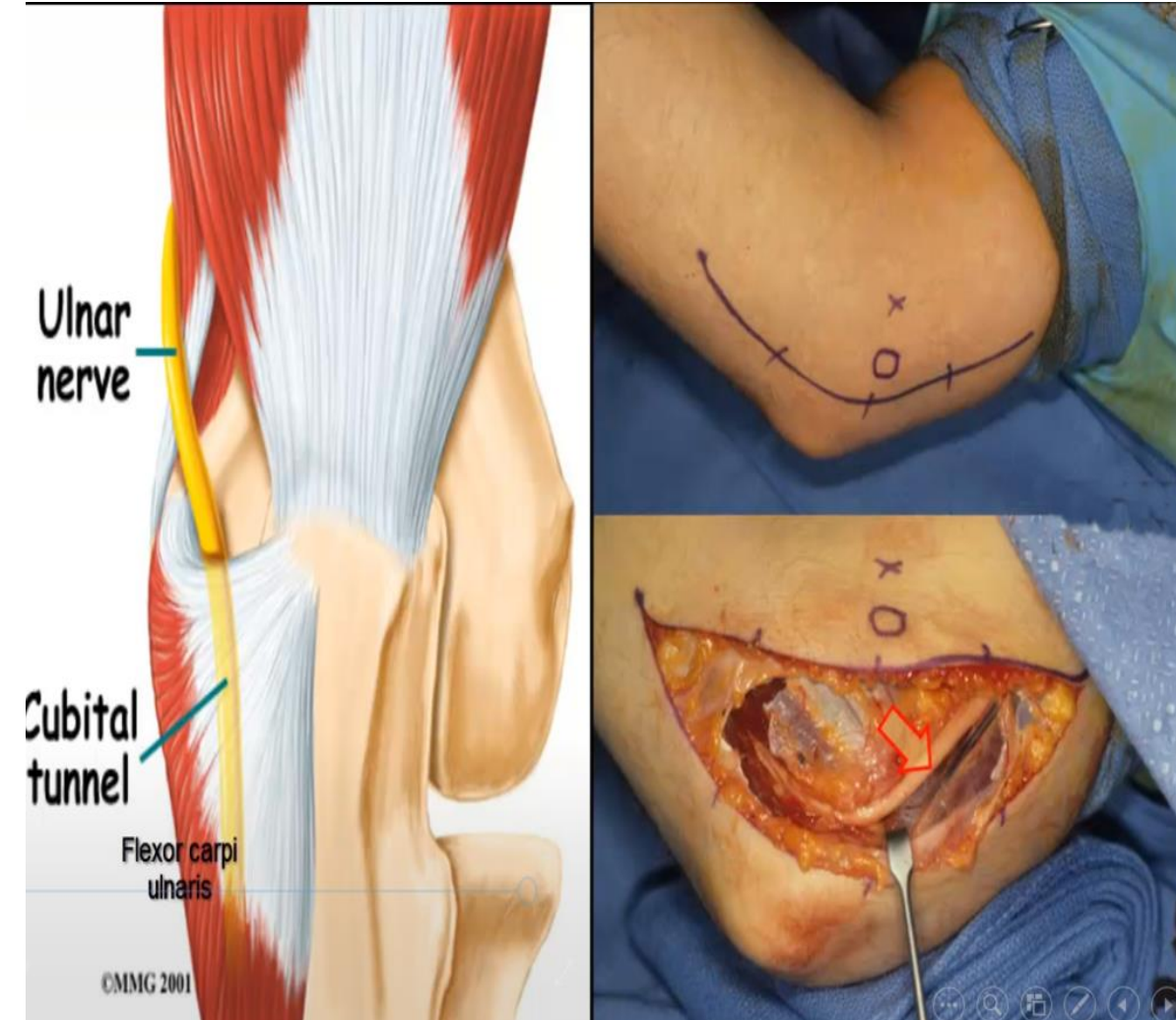
Medial Epicondylar Fracture

Q. Which clinical sign best indicates ulnar nerve injury in medial epicondylar fracture?

- a) Ape hand
- b) Partial clawing of 4th and 5th fingers
- c) Wrist drop

Q/ Which ligament attaches to the medial epicondyle and may be compromised in fracture?

- a) Annular ligament
- b) Ulnar collateral ligament
- c) Radial collateral ligament



Supracondylar fracture

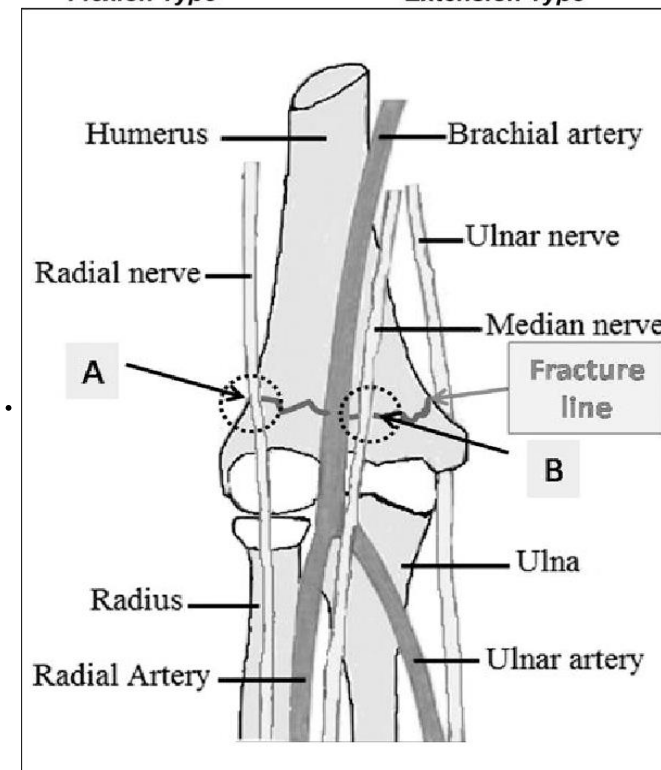
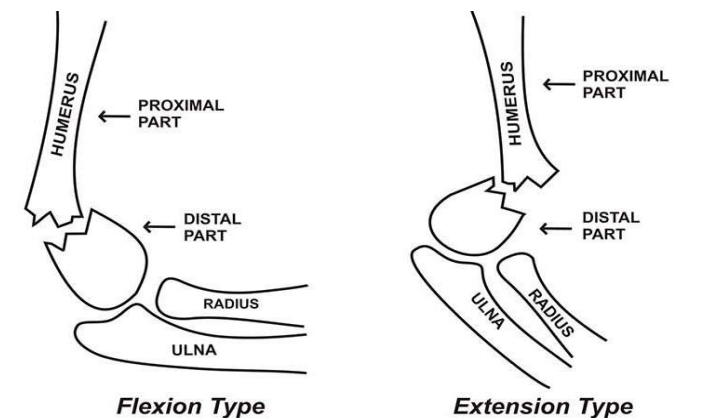
Pediatric Supracondylar Humerus Fractures

Case scenario: 6-year-old boy falls from monkey bars onto an **outstretched hand**, presenting with a swollen, **painful elbow held in extension**. Exam reveals no **radial pulse** but **warm, pink fingers**



Pediatric Supracondylar Humerus Fractures

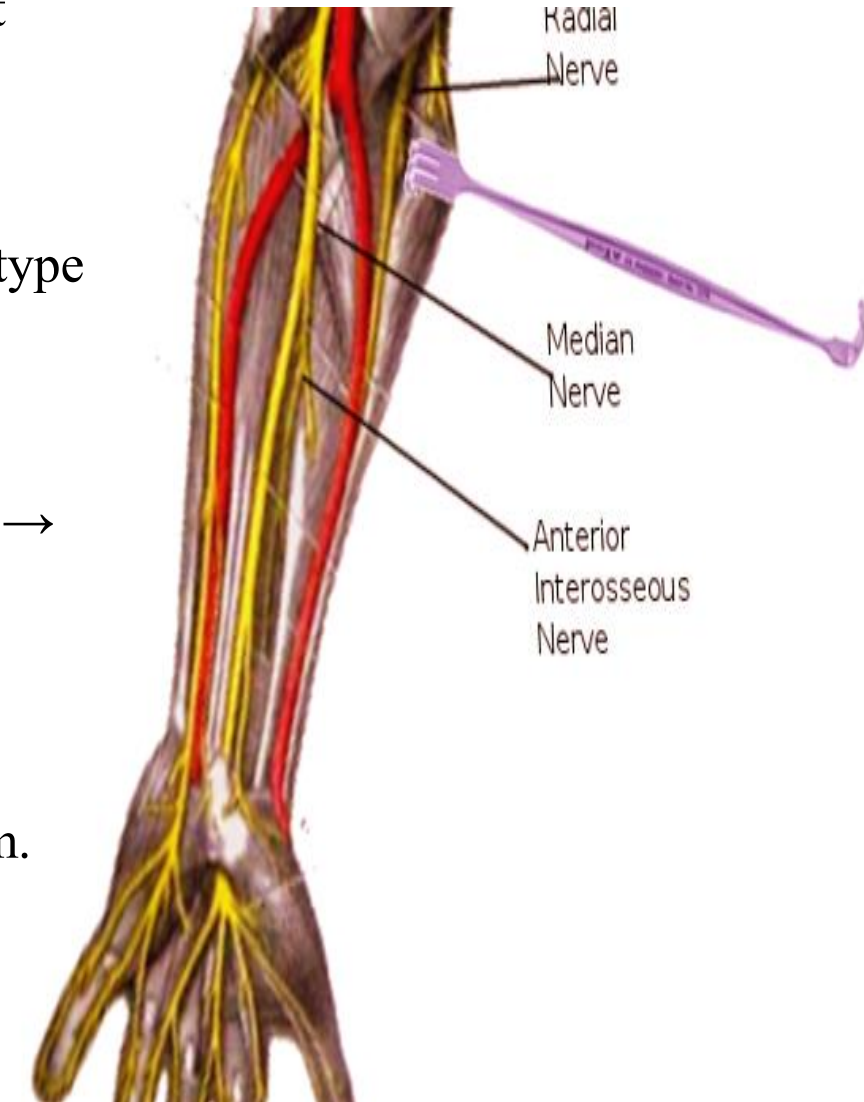
- **Definition:** Fracture of the distal humerus just above the elbow joint.
- **Age group:** Common in children 5–8 years.
- **Mechanism:** Fall on an outstretched hand (extension type most common).
- **Types:**
 - Extension type (>95%) – distal fragment displaced posteriorly.
 - Flexion type (<5%) – distal fragment displaced anteriorly.
- **Neurovascular structures at risk:**
 - **AIN (branch of median nerve):** most common in extension type.
 - **Radial nerve:** second most common.
 - **Brachial artery + Median nerve:** risk with posterolateral displacement.
 - **Complication:** Volkmann's ischemic contracture due to brachial artery injury.
 - **Ulnar nerve:** typically in flexion type.
 - **Key Exam:** Neurovascular assessment (pulse, capillary refill, Doppler if needed).



Pediatric Supracondylar Humerus Fractures

Anatomical Focus

- 1. Median nerve:** C6–T1, passes anterior to brachial artery, prone to injury at elbow → *Ape hand, Benediction hand*.
- 2. Anterior interosseous nerve (AIN):** pure motor branch of median nerve; palsy leads to inability to flex thumb IP and index DIP → “OK” sign failure.
- 3. Ulnar nerve:** C8–T1, passes behind medial epicondyle, injured in flexion type fractures → *claw hand*.
- 4. Radial nerve:** runs along posterior humerus, affected in posteromedial displacement.
- 5. Brachial artery:** at risk in extension type with anteromedial displacement → ischemia & Volkmann’s contracture.
- 6. Bony landmarks:** Relationship of olecranon + epicondyles remains intact (triangle preserved in fracture, lost in dislocation).
- 7. Radiographic lines:**
 - 1. Anterior humeral line** should pass through middle third of capitulum.
 - 2. Radiocapitellar line** should bisect capitulum.



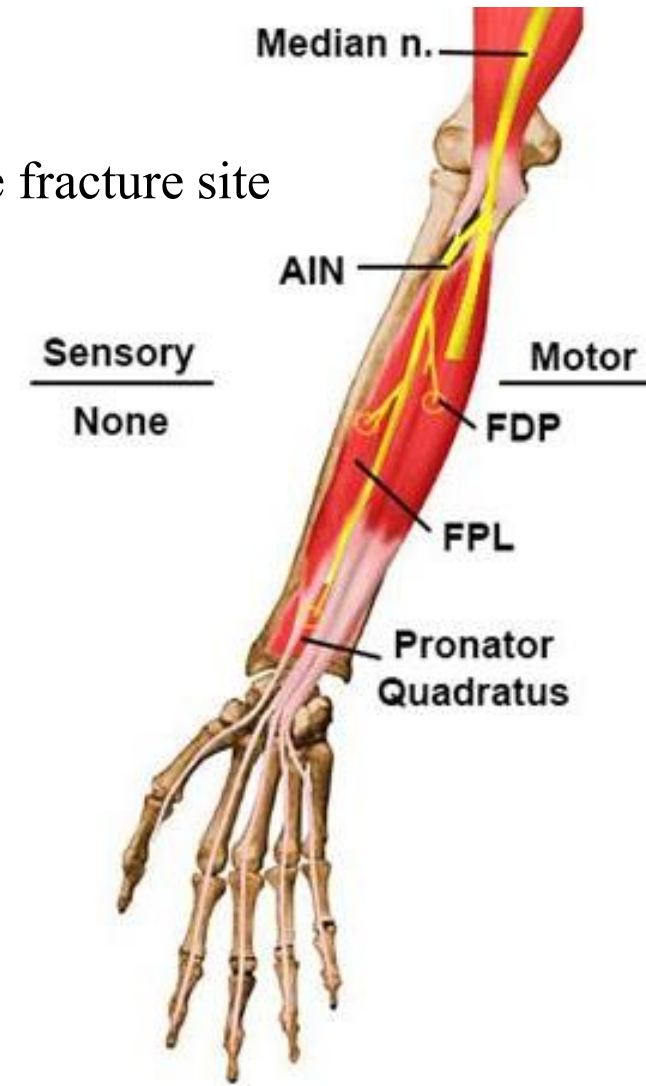
Pediatric Supracondylar Humerus Fractures

Q. Why does an extension-type supracondylar fracture more commonly injure the AIN than the ulnar nerve?

- a) The AIN provides sensory fibers to the palm
- b) The ulnar nerve is protected by the flexor carpi ulnaris muscle
- c) The AIN fascicles lie in the dorsal part of the median nerve, directly behind the fracture site
- d) The ulnar nerve crosses anterior to the distal humerus
- e) The AIN passes posterior to the medial epicondyle

Q. Which statement BEST describes the dual mechanism leading to AIN injury in supracondylar fracture?

- a) Stretching in Zone I and entrapment in carpal tunnel
- b) Compression of dorsal fascicles at fracture site and tethering in Zone II
- c) Entrapment between pronator teres heads and compression in Guyon's canal
- d) Direct trauma at medial epicondyle and superficial location in cubital fossa
- e) Avulsion due to pull of flexor digitorum profundus



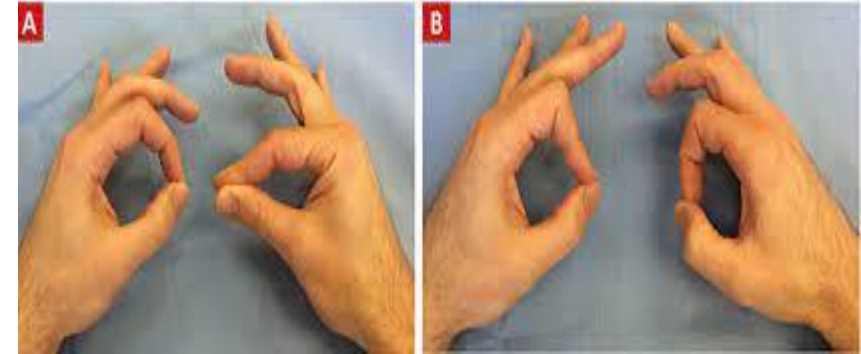
Pediatric Supracondylar Humerus Fractures

Which clinical test detects AIN palsy?

- a) Froment's sign
- b) "OK" sign
- c) Phalen's test
- d) Tinel's sign
- e) Cozen's test

Injury to brachial artery in supracondylar fracture most likely leads to:

- a) Erb's palsy
- b) Volkmann's ischemic contracture
- c) Saturday night palsy
- d) Popeye deformity
- e) Radial tunnel syndrome



❑ What is Volkmann's ischemic contracture?

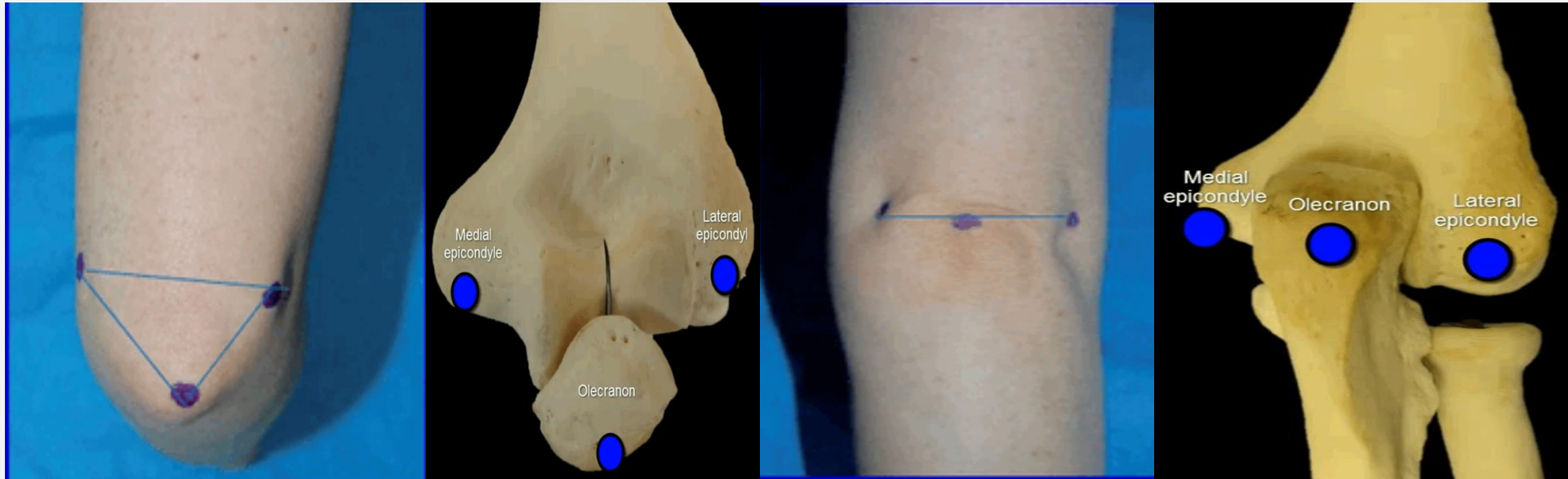
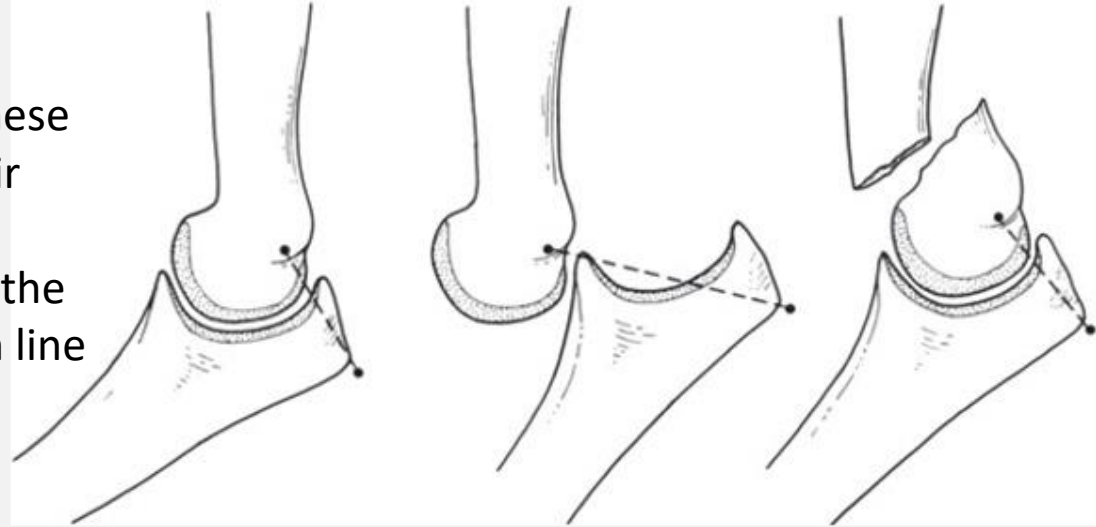
In supracondylar fracture

- There is displacement of proximal bone fragment forward & downward, compressing brachial a. & median n.
- When the collateral circulation cannot compensate for occlusion of the brachial artery leads to ischemia of flexor muscles of forearm & necrosis
- the muscles replaced by fibrous tissue leading to thinning & shortening of the muscle
- result in flexion deformity of wrist & fingers
(Volkmann's contracture)



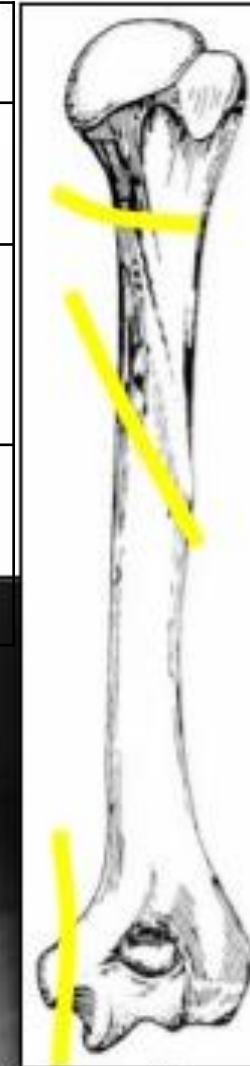
□ why in supracondylar fracture of humerus ,triangle relationship of the 3 bony points(olecranon, medial & lateral epicondyles) is not disturbed but in elbow dislocation , it is disturbed?

- A supracondylar fracture lies above these points, which therefore remain in their triangular relationship to each other
- in dislocation of the elbow, however, the olecranon comes to lie more or less in line with the epicondyles



Fractures of the humerus

nerve	Site of contact	deformity
Axillary	Around surgical neck	Flat shoulder
Radial	Spiral groove	Wrist drop
ulnar	Behind medial epicondyle	Claw hand



Radiological anatomy of elbow joint

Bony Landmarks of the Elbow Joint

1. Medial and Lateral Epicondyles

2. Trochlea and Capitulum

3. Olecranon Process

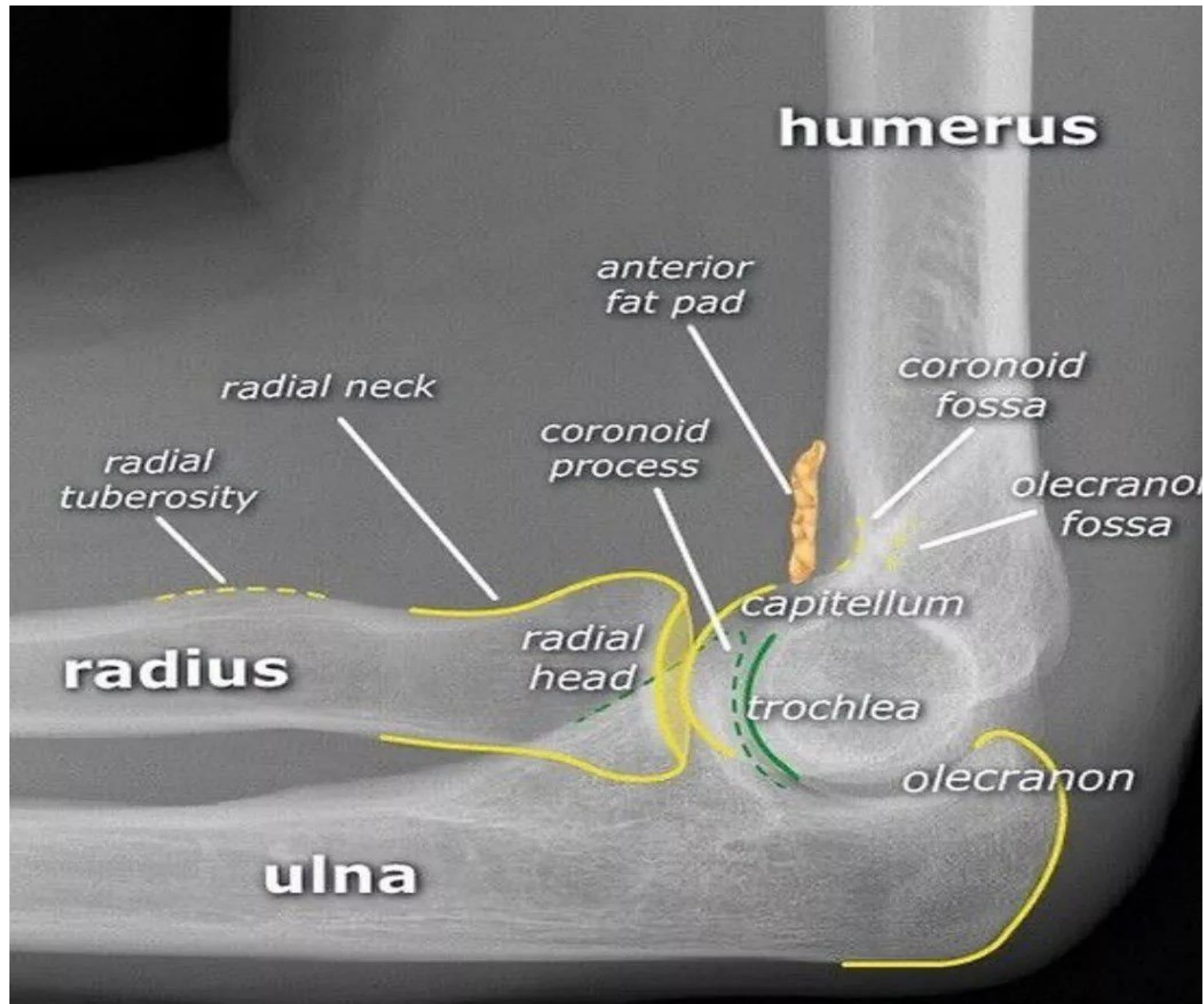
4. **Coronoid Process**/ Provides anterior stability to the elbow joint; fractures can lead to elbow instability

5. **Radial Head**/ Allows for forearm rotation; fractures can impair supination and pronation

6. **Radial tuberosity** / attachment of biceps tendon

7. **Trochlear notch**

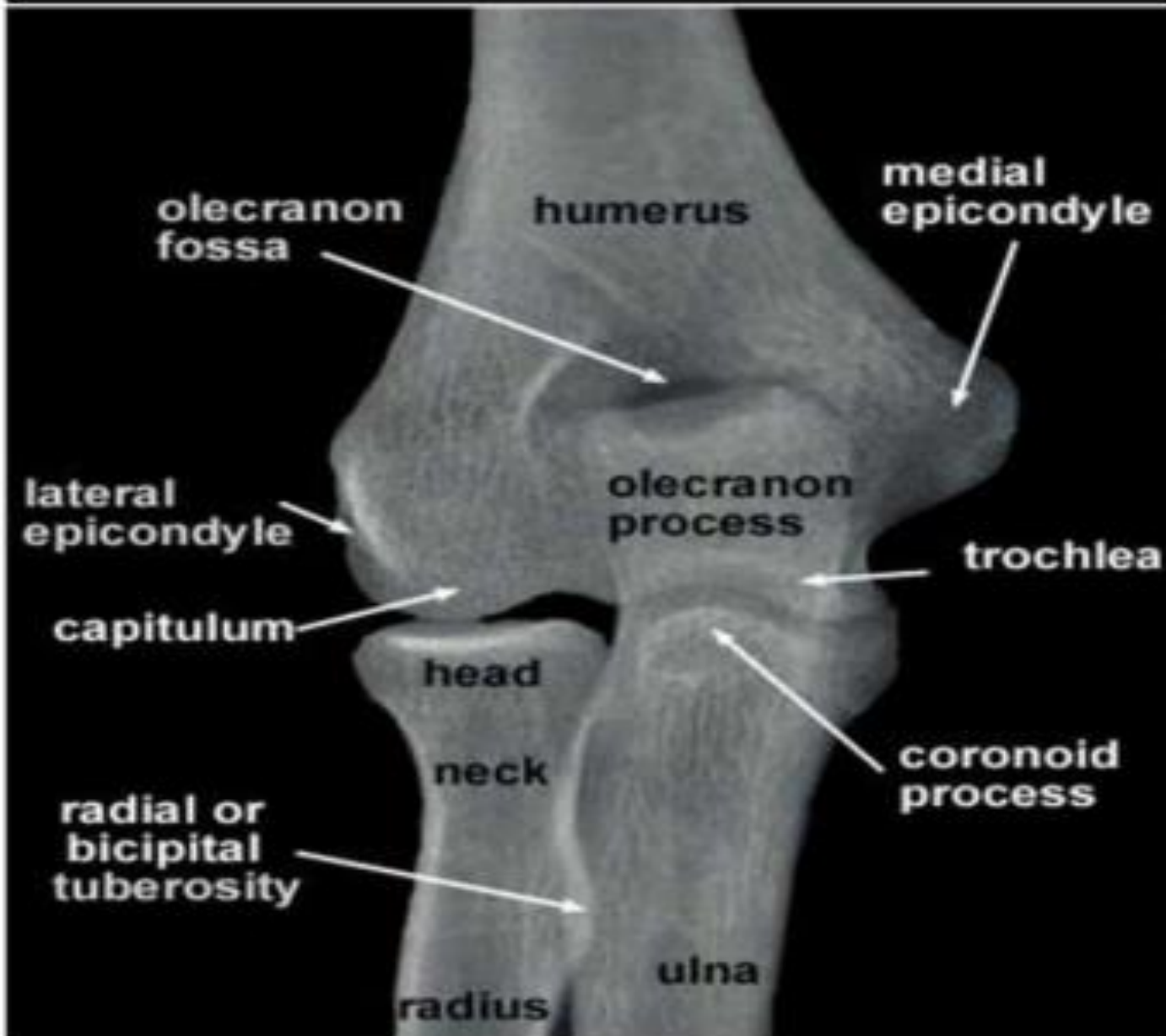
8. **Radial notch**



Centers of Ossification

- **Definition:** site of ossification
- **Types:**
 - **Primary Ossification Centers:**
 - Form the **shafts (diaphyses)** of long bones
 - Form the **bodies** of irregular bones
 - **Secondary Ossification Centers:**
 - Form the **ends (epiphyses)** of long bones
 - Usually appear **after birth**
 - Contribute to **growth & maturation** of skeleton
- **Radiological Importance**
 - **Age Estimation**
 - Predictable sequence of appearance & fusion
 - Used to assess **skeletal maturity** (pediatrics & forensics)
 - **Injury Diagnosis**
 - Secondary centers may mimic fractures on X-ray
 - Important to know **normal appearance & location**
 - **Monitoring Growth**
 - Radiographs help track **growth plate development**
 - Detects **abnormalities or disruptions** → possible deformities

Adult Elbow



Ped Elbow



Centers of Ossification in the Elbow

These are the usual 6 secondary ossification centers around the pediatric elbow, in order of appearance: **(CRITOE)**

- | | |
|--------------------------------------|---------|
| 1. C — Capitellum | 1 year |
| 2. R — Radial head | 3 year |
| 3. I — Internal (Medial) epicondyle | 5 year |
| 4. T — Trochlea | 7 year |
| 5. O — Olecranon | 9 year |
| 6. E — External (Lateral) epicondyle | 11 year |



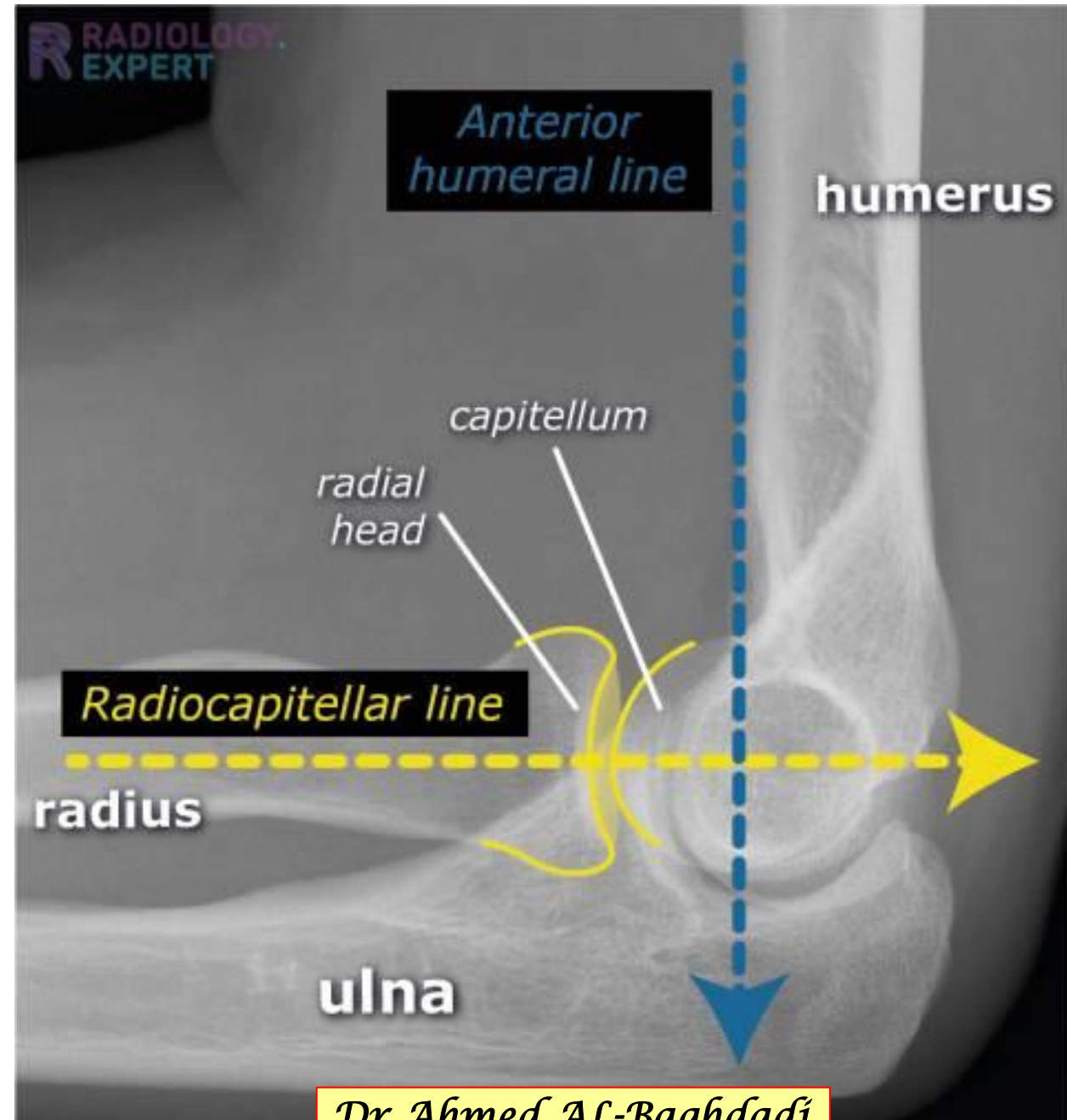
Radiological Lines and Their Clinical Relevance

1. Anterior Humeral Line (AHL)

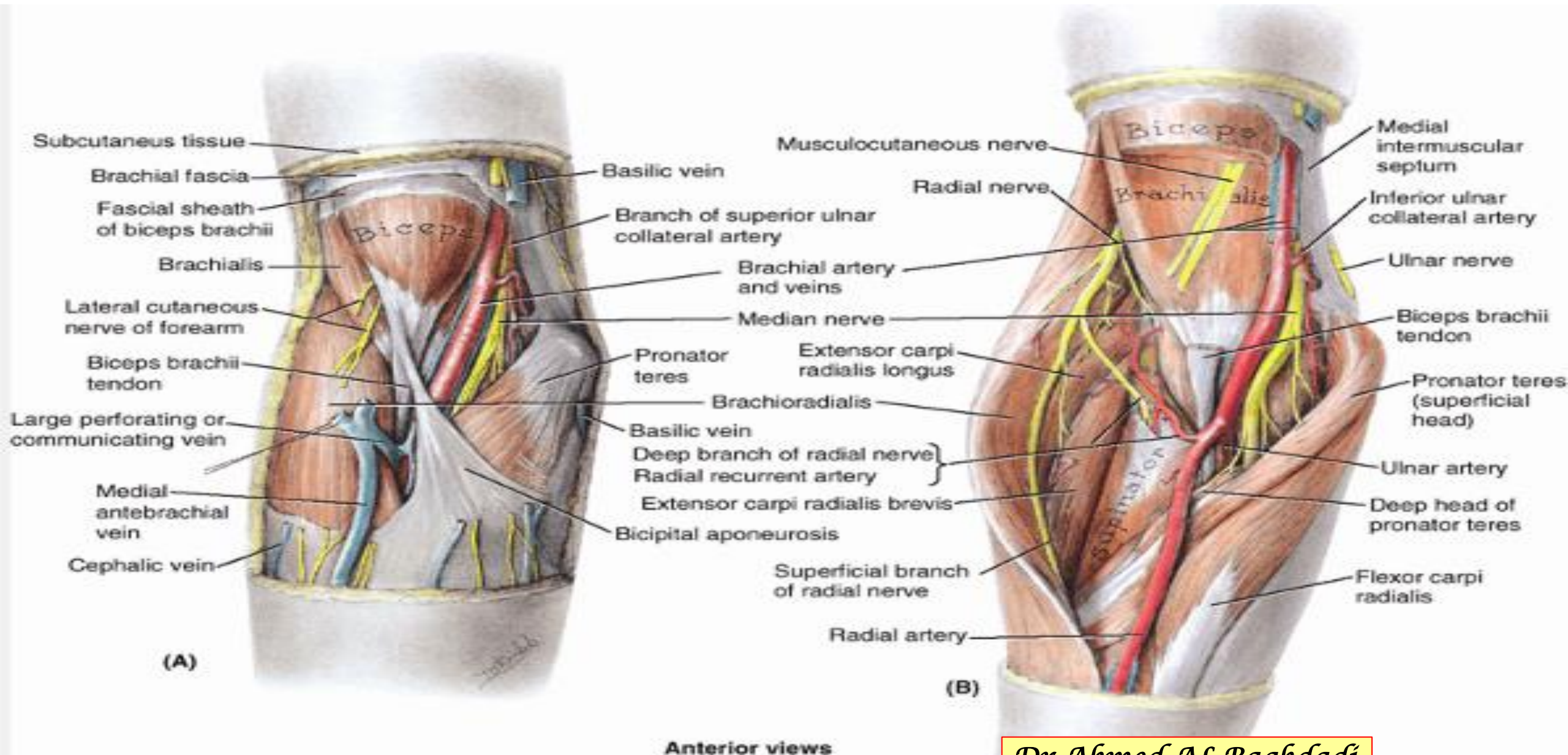
- a) **Description:** A line drawn along the anterior cortex of the distal humerus.
- b) **Clinical Importance:** In a normal elbow, this line should intersect the middle third of the capitellum. Displacement indicates potential supracondylar fractures

Radiocapitellar Line

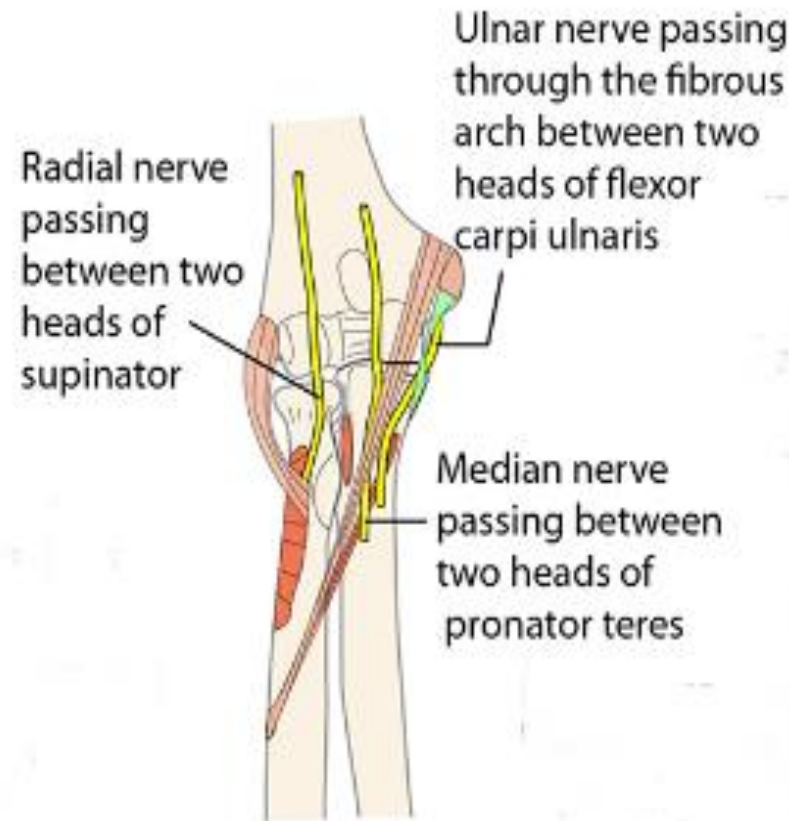
- a) **Description:** A line drawn through the center of the radial head and neck.
- b) **Clinical Importance:** In a normal elbow, this line should pass through the center of the capitellum. Displacement suggests radial head dislocation or other joint abnormalities.



Relations of nerves & arteries in the elbow region



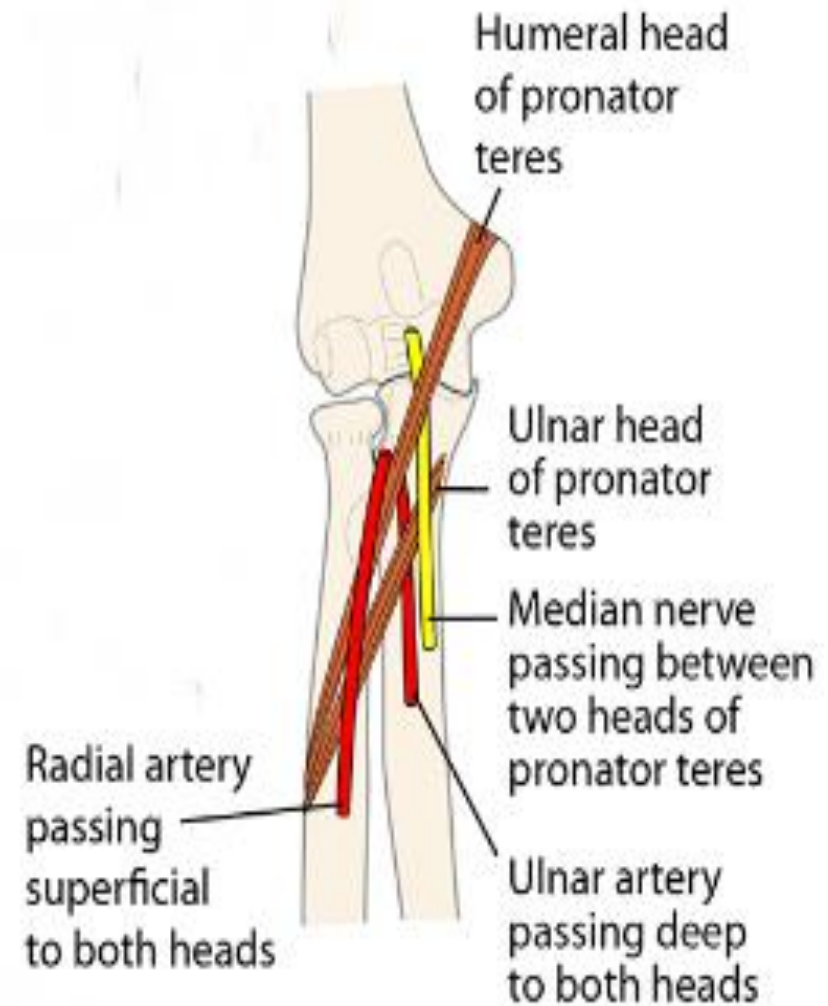
Relations of nerves & arteries in the elbow region



RULE 1

Nerves never pass between bones but pass around them.

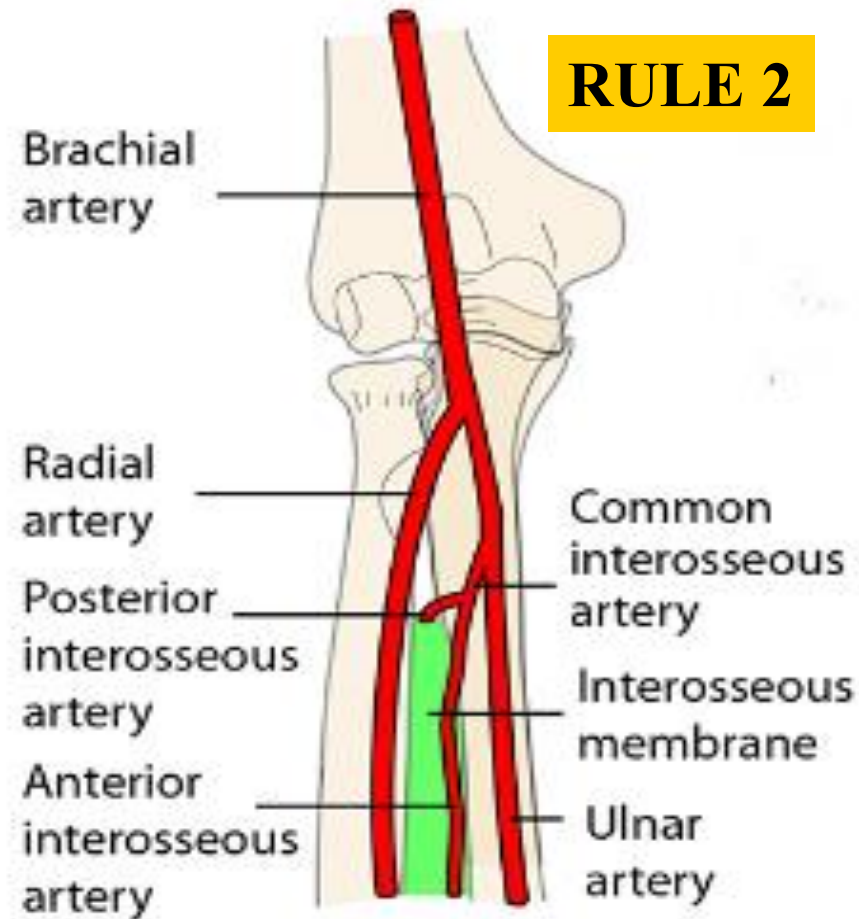
Radial, median and ulnar nerves all pass between the two heads of muscles



Relations of the two heads of pronator teres

Dr. Ahmed AL-Baghdadi

Relations of nerves & arteries in the elbow region



Arteries may pass between two bones as here where the posterior interosseous artery passes between ulna and radius

Centers of Ossification in the Elbow

Q. A 3-year-old child is imaged after elbow trauma. The only ossification center visible is the capitellum. Which other center is expected to appear next in sequence?

- a) Radial head
- b) Trochlea
- c) Lateral epicondyle

Answer: a) Olecranon 

Explanation: Olecranon ossifies around age 9; awareness prevents misdiagnosis as fracture.

Centers of Ossification in the Elbow

Q. During evaluation of a 9-year-old with elbow pain, the X-ray shows an ossification center posteriorly, which could be mistaken for an avulsion injury. Which center is this?

- a) Olecranon
- b) Capitellum
- c) Trochlea

Answer: a) Radial head 

Explanation: Sequence follows CRITOE: Capitellum (1 yr) → Radial head (3 yrs).

Radial Head Subluxation (Nursemaid's Elbow)

Radial Head Subluxation (Nursemaid's Elbow)

Case scenario: A 2-year-old girl presents with **sudden refusal to use her left arm** after her mother **pulled her up by the wrist**. The arm is held in **slight flexion and pronation** with **no swelling or deformity**



Radial Head Subluxation (Nursemaid's Elbow)

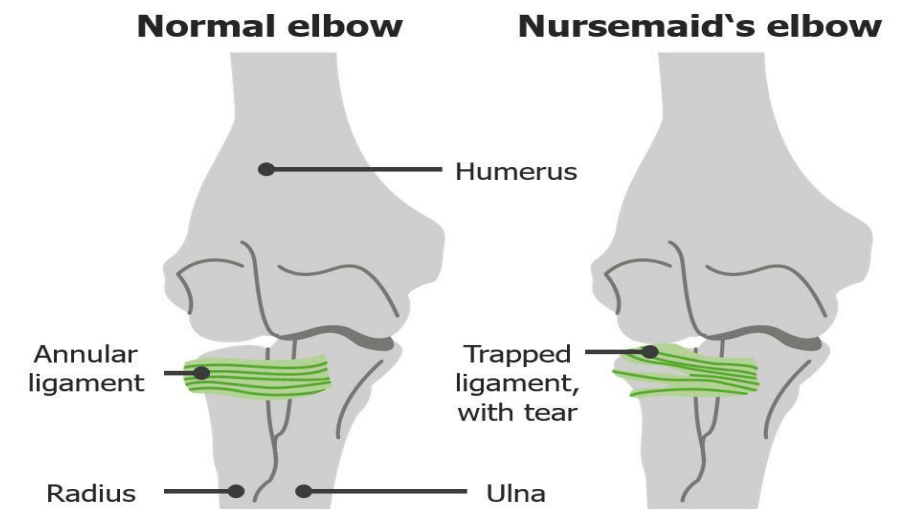
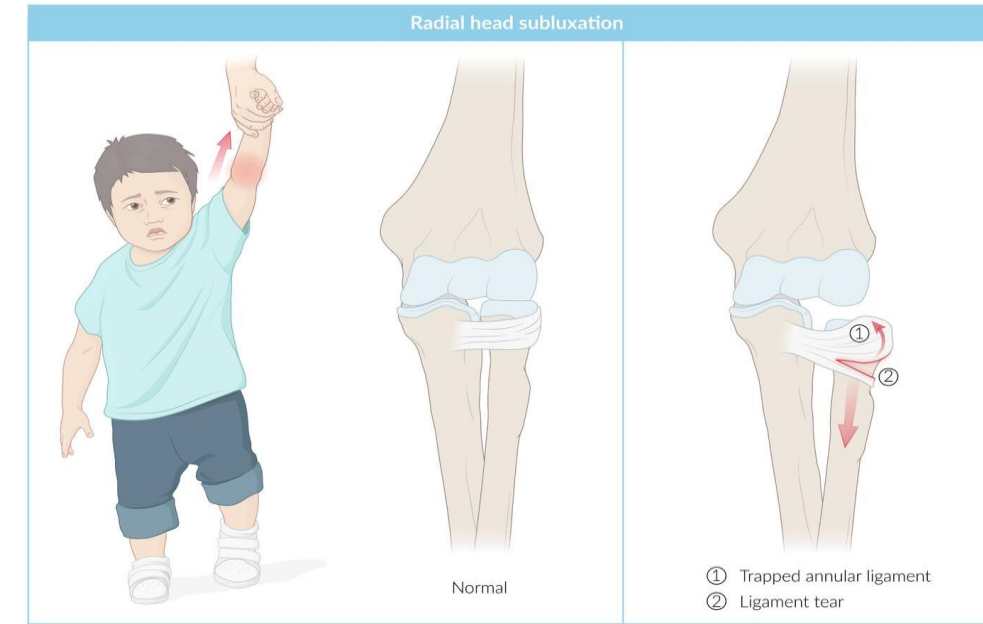
Clinical Keys

- **Definition:** Subluxation of the radial head from under the annular ligament.
- **Age group:** Most common in children **1–4 years** (ligaments are lax, radial head underdeveloped).
- **Mechanism:** Sudden longitudinal traction on a pronated, extended forearm (e.g., pulling child by the hand).

annular ligament slips over radial head → trapped in joint

(radiocapitellar joint)

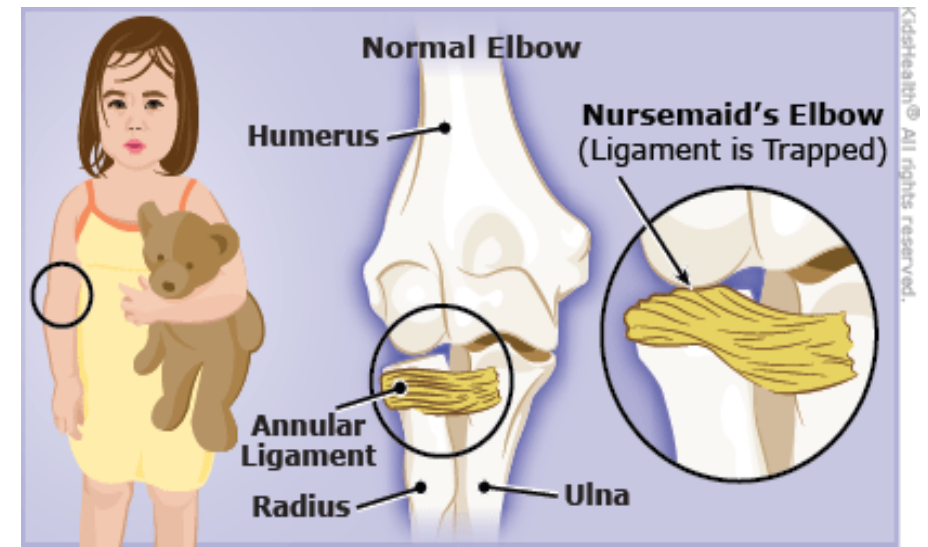
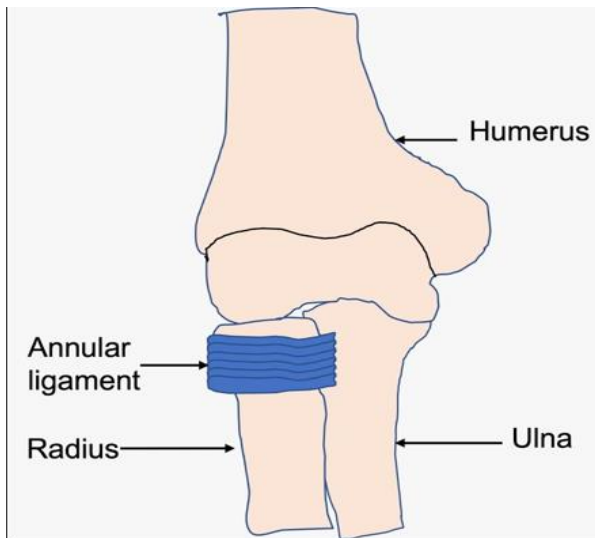
- **Typical position:** Child refuses supination, holds arm pronated + partially flexed, no swelling or bruising.
- **Management:** Closed reduction → supination + flexion (click felt). No immobilization usually required.



Radial Head Subluxation (Nursemaid's Elbow)

Anatomical Focus

- **Radial head:** Small, cylindrical, not fully developed in young children → easily slips.
- **Annular ligament:** Encircles radial head, holds it against ulna. In pulled elbow, ligament slips and gets trapped between radial head & capitulum.
- **Proximal radioulnar joint:** Site of subluxation, functionally affected in pronation-supination.
- **Growth factor:** Radial head enlarges with age, making recurrence less likely after ~5 years



Radial Head Subluxation (Nursemaid's Elbow)

Q. The most important anatomical structure involved in nursemaid's elbow is:

- a) Biceps tendon
- b) Annular ligament
- c) Radial collateral ligament
- d) Ulnar collateral ligament
- e) Interosseous membrane

.

Answer:  b

Explanation: The annular ligament slips and gets trapped → prevents radial head rotation

Radial Head Subluxation (Nursemaid's Elbow)

Q. A child with nursemaid's elbow typically holds arm in:

- a) Supination, full extension
- b) Flexion, pronation
- c) Abduction, external rotation
- d) Hyperextension
- e) Neutral with wrist drop

.

Answer:  b

Explanation: Child refuses supination, arm held pronated and flexed.